

COI - AI3 - Descriptive Written Test

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Answers

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① Smart Home Automation System

The intelligent agent in smart home will receive i/p from motion sensors, temperature sensors and user devices to automate the household.

→ Simple Reflex Agent

This agent makes decision based on the current environment.

Example:-
* If motion detected - Turn ON light
* If no motion detected - Turn OFF.

Advantage

- * Very fast response
- * Easy to implement

Disadvantage

- * Can't remember previous events
- * Can't learn user habits.

→ Model-Based Reflex Agent

It maintains an internal model or memory of the environment.

Example: It remembers whether lights are already ON or OFF

Advantage

- * Better at decision making
- * Reduce unnecessary options.

Disadvantage.

- * More complex
- * More memory

→ Goal-Based Agent

It chooses actions that help achieve the pre-defined goals.

Example:- If temperature is 30°C , it

- opens windows
- turn on fan
- start ac.

Advantage

- * Flexible
- * Better planning

Disadvantage:

- * slower
- * Requires planning.

→ Utility - Base Agent

It selects the action which has the highest overall benefits.

Example: * At night, keeps bedroom warm

* locks door automatically

* check CCTV's.

Advantages

* Balances comfort

* learns about user

Disadvantages.

* Intensive

* Requires more data.

Most Effective Approach.

A hybrid Agent which combines Model-Based, Goal-Based and Utility Because:-

* Model-Based - remembers previous state

* Goal-Based - ensure comfort & safety.

* Utility-Based - selects the best action in overall.

③ Robot placed in unknown Haze.

It must observe its surroundings, where to move and reach the exit without the map. It uses searching algorithm to explore.

Uniform cost search (UCS)

It will expand the path with lowest cumulative cost & ensures least-cost path.

Example: path costs:

straight path = 1

Rocky path = 5

slippery path = 3

* The robot chooses straight path as it has the lowest path cost

Advantages

- * Finds optimal path
- * Complete & Reliable

Disadvantages

- * High memory
- * Slow for large maze

Complexity

→ Time: $O(b^{b^* / \epsilon})$

→ complexity: $O(b^{b^* / \epsilon})$

b → branching factor

b^* → solution cost

ϵ → min. step cost

Iterative Deepening Search (IDS)

It repeatedly performs Depth-First Search with increasing depth limits.

Example: Depth 0 $\rightarrow$ explores start node
Depth 1 $\rightarrow$ explores level 1
continues till exit is found.

Advantages

- * Less memory
- * Complete

Disadvantages

- * Repetition of upper levels

Complexity

$\rightarrow$ Time: $O(b^d)$

$\rightarrow$ Space: $O(bd)$

$b \rightarrow$ branching factor

$d \rightarrow$ depth of solution

Relation to Intelligent Agents.

* Simple Reflex Agent

- Moves based on current observation only
- Poor for unknown maze

* Model-Based Agent

- Maintains internal map
- Avoids revisiting paths.

* Goal-Based Agent

- plans the path towards exit using search algorithms.

* Utility-Based Agent

- chooses path by distance, safety and movement cost.

Best combination

- * Goal-Based Agent with uniform cost search

Because:-

- Goal-Based focus on reaching exit
- UCS guarantees lowest cost path
- Robot avoids risky routes.
- suitable for unknown environment.

- * If memory is limited, Model-Based Agent with Iterative Deepening Search (IDS) is good because IDS requires much less memory & model helps to avoid repeated mistakes.